

MODELING THE DIAMETER DISTRIBUTION ON THE FULLY STOCKED HARDWOOD PERMANENT SAMPLE PLOTS

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ABSTRACT.—The purpose of this work was to develop a diameter distribution model that adequately represents the observed data from Nova Scotia's fully stocked research Permanent Sample Plots (PSP). We have been able to develop a family of Weibull distributions whose parameters depend only on the stand average diameter. Using both chi-square and Kolmogorov Smirnov tests, we tested the hypothesis that the distribution of diameters observed on individual plots is from that member of this family of distributions whose parameters correspond to the plot mean diameter. At the 99 percent level we would not reject this hypothesis on all except 10 of the 283 plot samples. At the more stringent 95 percent level, we reject this hypothesis only for 32 out of the 283 plots.

INTRODUCTION

This work came about while completing the normal yield tables for Nova Scotia hardwoods since we wanted to have a quantitative description of the diameter distribution of the trees making up a stand. In the older softwood normal yield tables (Nova Scotia Department of Land and Forests 1990), this distribution of diameters is given as three numbers, namely the volume available in diameters larger than 1.5, 9.1, and 14.1 cm diameter at breast height (d.b.h.). These indicate total volume plus merchantability limits of two classes of products. The problem with this approach is that critical diameters for merchantability change depending on harvest technology being used. Another reason for investigating the development of a diameter distribution is to enhance the capabilities of the Nova Scotia Geographic Information System (GIS). The GIS data will include only a single stand average diameter. The ability to project a diameter distribution from this single number will be useful in developing quantitative assessments of the forest cover and of its economic value.

There is a long history of using the Weibull distribution to model diameter distributions (Clutter et al. 1983, Bailey and Dell 1973). Obviously, this influenced us to use the Weibull as our distribution of choice. The Weibull distribution requires the estimation of three parameters. For any given stand, it is not difficult to estimate the parameters that best describe that stand. A standard technique involves Maximum Likelihood Estimators (MLE). Law and Kelton (1989) discuss parameter estimation in general and give methods for several standard distributions, including the Weibull. The derivation of MLE's for the Weibull was originally given by Harter and Moore (1965). Bailey and Dell discuss various approaches taken to estimation of these parameters for forest stands.

Our problem is unusual in that we do not want to estimate the parameters for any particular stand. Rather, we want to develop a continuous family of distributions that depend only on the single variable of stand average diameter. This paper thus attempts to answer two questions. First, how can we develop the parameters of the Weibull distribution as a function of stand average diameter? Second, does the family of distributions do a reasonable job of describing the distribution of diameters when applied to any particular stand? As will become evident, our approach is heuristic in two aspects. First, most natural stands have an understory with trees of very small diameter. These are often large in number but insignificant in basal area. If these are included in the diameter distribution, this will almost inevitably mean a bimodal distribution. We developed a screening mechanism designed to filter these from our data base. Second, fitting the location parameter for a Weibull is somewhat problematic. Our choice here has been heuristic and is open to criticism. However, it appears to work well and we will argue that it does not significantly influence the results.

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Briefly, our approach has been to first group the data from our plots into diameter class "superplots". We then carry out MLE estimates of the parameters for each superplot and attempt to fit smooth curves to these parameter estimates using the average diameter of the superplot as the variable to be regressed against the parameter estimates. It turns out that the curves we fit are quite reasonable approximations to the MLE estimates. The correspondence between the superplot histograms and the density functions is very strong. Even more encouraging, if we apply the Weibull functions to the individual plots, a very large portion of the plots give good hypothesis testing results.

We originally expected that the diameter distributions of treated (thinned) stands would be greatly different from the natural stands that we used for the normal yield tables. We were surprised to find that this was not the case. The results of hypothesis testing using the Weibull distributions on the entire set of research permanent sample plots (PSP's) gave results very comparable to tests restricted to the fully stocked control plots that we use for the normal yield tables.

OUR PROCEDURE

The data was taken from Nova Scotia's PSP's. There are 80 plots identified originally as the fully stocked stands used to represent the normal yield. Most were first measured in 1978 and remeasured every 5 years since. We have a total of 283 plot measurements. As is common practice, we treat the remeasurement data as independent samples.

Our approach is to first group the data from the individual PSP's according to plot average diameter ($\bar{d}$). We used 1 cm diameter classes. It is important to clarify what we mean by the average diameter $\bar{d}$ of the trees on the plot.

The $\bar{d}$ is the diameter of the tree of average basal area on the plot. Thus if we have sample diameters from n trees on the plot ($d_1, d_2, \dots, d_n$), then :

$$(1) \quad \bar{d} = \sqrt{\frac{1}{n} \sum_{i=1}^n d_i^2}$$

The Weibull distribution has a simple analytic form. There are three parameters, commonly referred to as the location (λ), scale (β) and shape (α) parameters. Using these parameters, if the random variable D , the diameter of a tree on a plot, has a Weibull density function $f(D)$, then

$$f(D) = \begin{cases} \alpha \beta^{-\alpha} (D - \lambda)^{\alpha-1} e^{-((D-\lambda)/\beta)^\alpha} & \text{if } D > \lambda \\ 0 & \text{otherwise} \end{cases}$$

and the corresponding cumulative distribution function is $F(D)$ where

$$F(D) = \begin{cases} 1 - e^{-((D-\lambda)/\beta)^\alpha} & \text{if } D > \lambda \\ 0 & \text{otherwise.} \end{cases}$$

As indicated earlier, we set out to find an expression for α , β , and λ as a function of stand average diameter and to then test the performance of this family of distributions.

The Insignificant Basal Area Test

Almost every plot contains a number of small trees that are unlikely to become crop trees because they are undergrowing the main stand. This may correspond either to ingrowth or to older trees that have always been suppressed in the stand. A quite typical situation is small-diameter American beech undergrowing larger-diameter sugar maple. These trees may be quite numerous but make up an insignificant fraction of the basal area of the main stand. If we take the time to examine the trees on the plot manually, it is usually not too hard to decide which trees

have no potential as crop trees . However, we need an automated procedure for routine processing. We developed a screening test that has proven to be quite effective. Our approach is based on the idea of a basal-area-weighted

average diameter ($\bar{d}$). The calculation is : $\bar{d} = \frac{\sum_{i=1,n} d_i \frac{d_i^2}{\sum_{j=1,n} d_j^2}}{\sum_{i=1,n} \frac{d_i^2}{\sum_{j=1,n} d_j^2}} = \frac{\sum_{i=1,n} d_i^3}{\sum_{j=1,n} d_j^2}$.

The idea is to delete trees that have an insignificant effect on this basal-area-weighted diameter. The test quantity

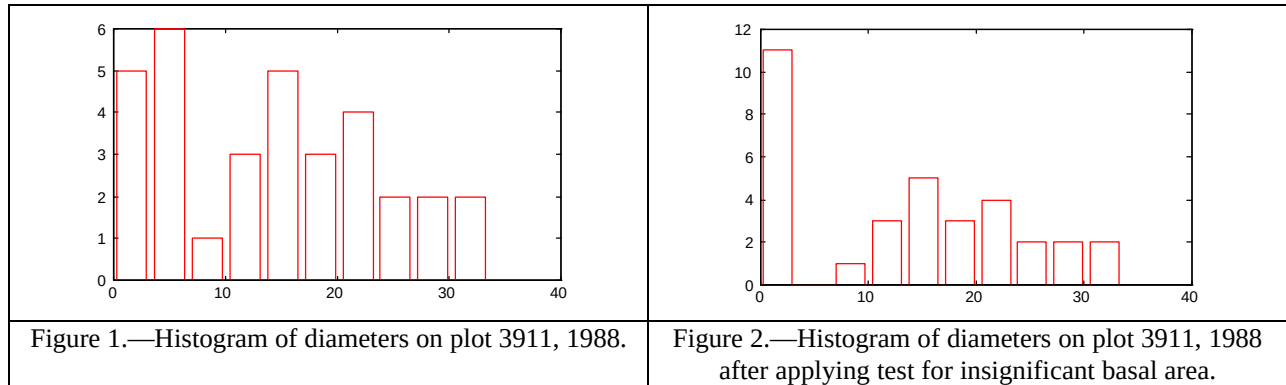
we use is $d_{\text{test}} = .10 \bar{d} / n$ and we discard a tree if $\frac{d_i^3}{\sum_{j=1,n} d_j^2} < d_{\text{test}}$.

The motivation of this test is that, since $n d_{\text{test}} = .1 \bar{d}$, it is impossible to discard enough trees to change this basal-area-weighted diameter by more than 10 percent .

Using the example plot 3911, $\sum_{j=1,n} d_j^2 = 9925.2$, and there are 28 trees with positive diameter for an average

diameter $\bar{d} = 18.5$ (Φιγυρε 1). The $d_{\text{test}} = .0736$ so that we discard any trees with have $d_i^3 < .0736 * 9925.2 = 730.49$, which corresponds to $d_i < 9.006$ cm. Applying this test to the plot 3911, we get the Figure 2 histogram where the left-hand column now contains both the dead and discarded trees. The average diameter of the remaining trees is $\bar{d} = 21.12$ cm.

Plot 3911 in 1988 illustrates the problem. As indicated in Figure 1, there are 5 dead trees on the plot. The larger trees are all sugar maple. There are three beech with diameters less than 4 cm and three white spruce with diameters from 4-6 cm. The next smallest tree has a diameter of 9.3 cm.



We applied this discarding procedure for trees of insignificant basal area to all of the plots., Trees were discarded for 106 of the 283 plots. Only 13 plots were problems in the sense that the deleted trees accounted for more than 5 percent of basal area. In no case did the discarded trees account for more than 7 percent of the total basal area of the plot even though, in some cases, more than half of the total trees were discarded (e.g., plot 4111 in 1978) . This procedure is simple and quite effective.

Grouping into Diameter Classes

After deleting the trees of insignificant basal area from each plot, the average diameter $\bar{d}$ was calculated for the plot, and all the remaining trees on the plot were added into the diameter class corresponding to the plot average diameter $\bar{d}$. As indicated earlier, we used 1 cm diameter classes for the aggregation. We computed the average and the minimum diameter in each diameter class. The results are shown in Table 1. We plotted the minimum diameter as a function of the average diameter of the d.b.h. class, the results are shown in Figure 3. Although a straight line represents the minimum diameter very well, we fit a regression line to a second-order equation. The least squares fit leads to the following function.

$$\text{Minimum diameter} = 0.4240 \bar{d} - 0.0020 \bar{d}^2$$

This function is plotted on Figure 3. The line below it is half of the minimum diameter line. A natural question arises as to how to deal with this minimum diameter. On the one hand, we could ignore it completely and try to fit Weibull distributions in which the location parameter $\lambda = 0$. However, since there is such a strong trend in the minimum diameter, we have decided to make use of this in developing a location parameter for the diameter distributions. Our approach has been simple in the extreme. We cannot use the curve of minimum diameter directly since this is only a least squares fit to sample data and some observations lie below this curve. The Weibull distribution implies that it is not possible to have any observations that lie below the location parameter. Our simple choice has been to use 75 percent of the minimum diameter line for our location parameter. As will be seen, this simple choice gives good results. However, it may be worthy of further investigation.

Table 1.—Statistics on Diameter Classes

d.b.h. Class	Number of Trees	Class Average Diameter	Class Minimum Diameter
<i>cm</i>		----- <i>cm</i> -----	
2-3	103	2.377	1.200
3-4	51	3.710	1.800
4-5	74	4.476	2.000
5-6	279	5.670	2.700
6-7	293	6.450	3.000
7-8	354	7.561	3.900
8-9	227	8.515	4.200
9-10	259	9.460	4.200
10-11	271	10.474	4.800
11-12	251	11.402	5.500
12-13	282	12.547	5.600
13-14	363	13.549	6.500
14-15	675	14.648	6.300
15-16	619	15.588	6.800
16-17	717	16.583	7.600
17-18	742	17.600	8.100
18-19	858	18.543	7.500
19-20	607	19.530	8.600
20-21	488	20.409	9.300
21-22	312	21.554	9.700
22-23	129	22.389	10.200
23-24	102	23.370	11.200
24-25	72	24.472	12.200
25-26	99	25.49	12.400
26-27	52	26.580	11.700
27-28	45	27.116	13.600
28-29	21	28.185	14.900
29-30	20	29.434	15.200

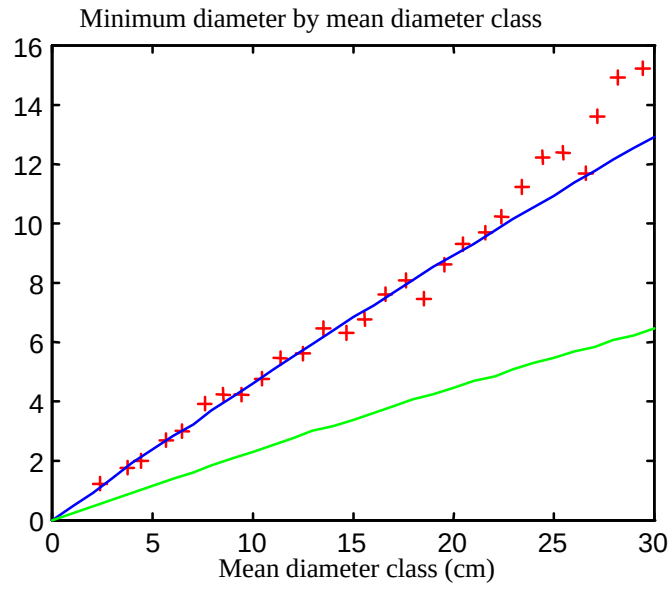


Figure 3.—Plot of minimum diameter as a function of class average diameter.

Table 2.—MLE Estimates of the Shape and Scale Parameters

d.b.h. Class	Number of Observations	Average Diameter	Location Parameter (λ)	Shape Parameter(α)	Scale Parameter (β)
<i>cm</i>		<i>cm</i>			
2	103	2.387	0.751	2.95248	1.75791
3	51	3.710	1.159	2.24311	2.69253
4	74	4.488	1.397	2.81771	3.31149
5	279	5.675	1.756	2.21469	4.12462
6	293	6.455	1.990	2.23072	4.70491
7	354	7.565	2.320	2.67054	5.60782
8	227	8.516	2.599	3.44816	6.36526
9	259	9.463	2.875	2.39152	6.98182
10	271	10.479	3.168	2.43799	7.76027
11	251	11.404	3.431	3.31249	8.57024
12	282	12.549	3.754	2.39408	9.31638
13	363	13.551	4.034	2.33548	10.05730
14	675	14.651	4.337	2.40972	10.92155
15	619	15.589	4.593	2.61117	11.71793
16	717	16.585	4.861	2.67427	12.50934
17	742	17.601	5.132	2.95346	13.36265
18	858	18.546	5.382	2.57950	14.00911
19	607	19.531	5.639	2.90072	14.87957
20	488	20.411	5.866	2.55067	15.47276
21	312	21.556	6.158	2.35983	16.28445
22	129	22.390	6.368	2.75982	17.11851
23	102	23.371	6.613	2.46647	17.78338
24	72	24.473	6.884	2.34868	18.60066
25	99	25.495	7.132	2.16640	19.22892
26	52	26.581	7.393	2.90041	20.50533
27	45	27.116	7.520	2.28428	20.64206
28	21	28.185	7.771	2.91684	21.85245
29	20	29.434	8.061	2.97601	22.88152

MLE Estimates of Shape and Scale Parameters

Once the location parameter is given, MLE estimates of the shape and scale parameters can be computed from equations originally developed by Harter and Moore. Law and Kelton give a simple Newton step iteration. This calculation was programmed in a FORTRAN program called Calcparms.f that is available from the authors. Convergence to four significant figures of accuracy is obtained in only three or four iterations.

Table 2 gives the results of the MLE estimates of α and β for each d.b.h. class. If we plot the β 's as a function of class average diameter, we see that a straight line fits them extremely well by (Figure 4):

$$\beta = -0.2959 + 7757 \bar{d}.$$

The behavior of the α 's is not as smooth as seen in Figure 5. A least squares fit to a quadratic model gives:

$$\alpha = 2.6744 - 0.0101 \bar{d} + 0.0003 \bar{d}^2.$$

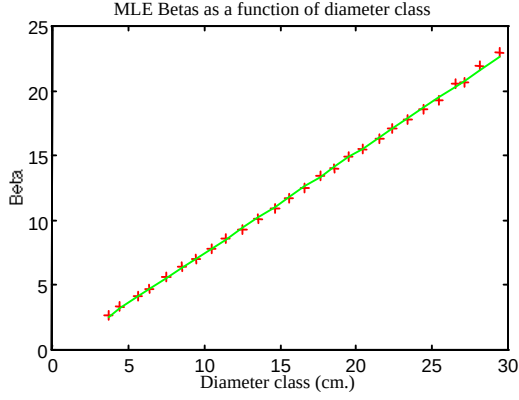


Figure 4.—MLE Beta s as a function of diameter class.

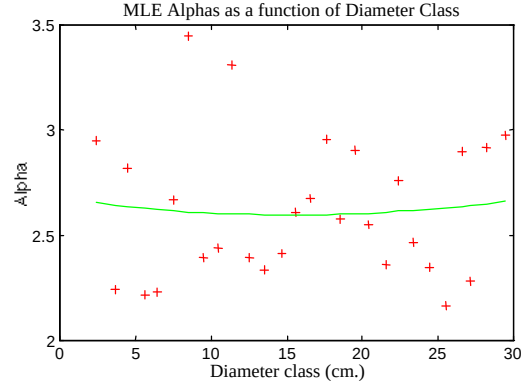


Figure 5.—MLA Alpha s as a function of diameter class.

If we use the MLE estimates for α and β directly, it turns out that even though β varies smoothly as a function of $\bar{d}$, the density functions do share this property. Figure 6 illustrates the problem where the transition between diameter distributions is not as smooth as it might be. This is due to the jumpiness of the MLE estimates of α .

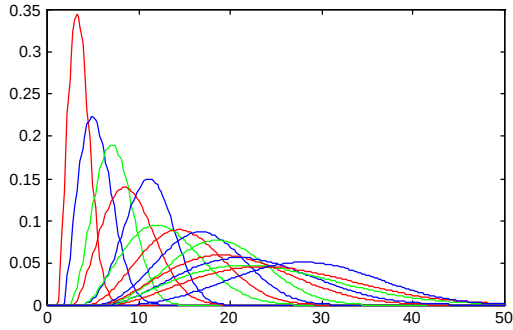


Figure 6.—Diameter distributions resulting from the MLE estimates of α and β .

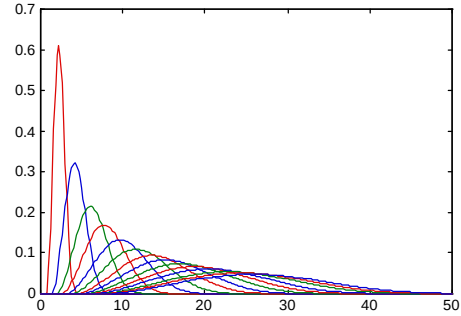


Figure 7.—Diameter distributions resulting from Smoothing the MLE estimates of α and β .

To complete our estimation process, we used the equation derived for α and reestimated the β . Once again, a straight line fits the β 's extremely well, but with slightly different parameters:

$$\beta = -0.2877 + 0.7784 \bar{d}$$

As seen in Figure 7, there is now a much smoother transition between diameter class distributions.

VERIFYING THE MODELS

The approach outlined results in a model of diameter distributions that can be summarized as:

Location Parameter

$$\lambda(\bar{d}) = 0.4240 \bar{d} - 0.0020 \bar{d}^2$$

Shape Parameter

$$\alpha(\bar{d}) = 2.6744 - 0.0101 \bar{d} + 0.0003 \bar{d}^2$$

Scale Parameter

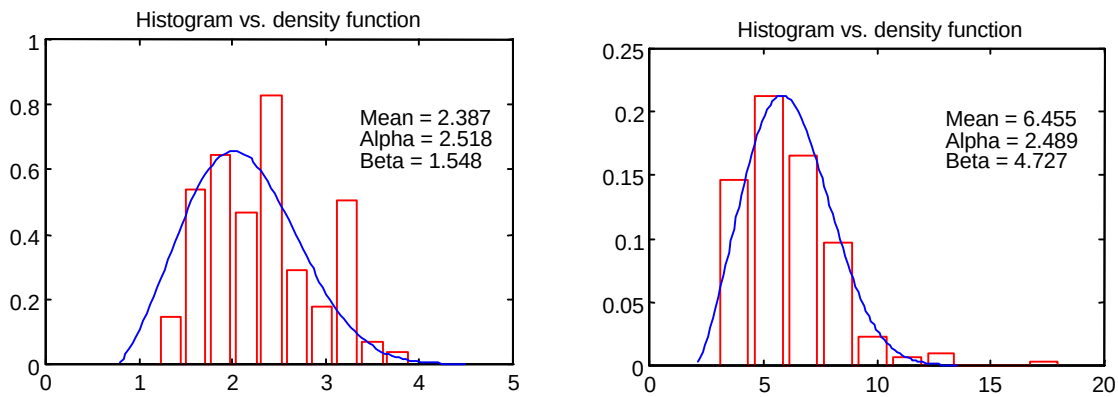
$$\beta(\bar{d}) = -0.2877 + 0.7784 \bar{d}$$

Using these parameters, the random variable $D(\bar{d})$, the diameter of a tree on a plot with average diameter $\bar{d}$, has a Weibull density function (all the parameters and variables are to be thought of as functions of $\bar{d}$):

$$f(D) = \begin{cases} \alpha \beta^{-\alpha} (D-\lambda)^{\alpha-1} e^{-((D-\lambda)/\beta)^\alpha} & \text{if } D > \lambda \\ 0 & \text{otherwise} \end{cases}$$

We have two approaches to verifying that our procedures have led to valid results. In Figure 8, we show the density functions versus a histogram of the data for several diameter classes. As is visually evident there, the fits are very good. Note that there are quite a few trees in these histograms. For example, as seen in Table 2, the 14 cm class has 675 trees.

This does not, however, answer the question as to how well the distributions represent any given plot. The approach that we have taken here is based on hypothesis testing.



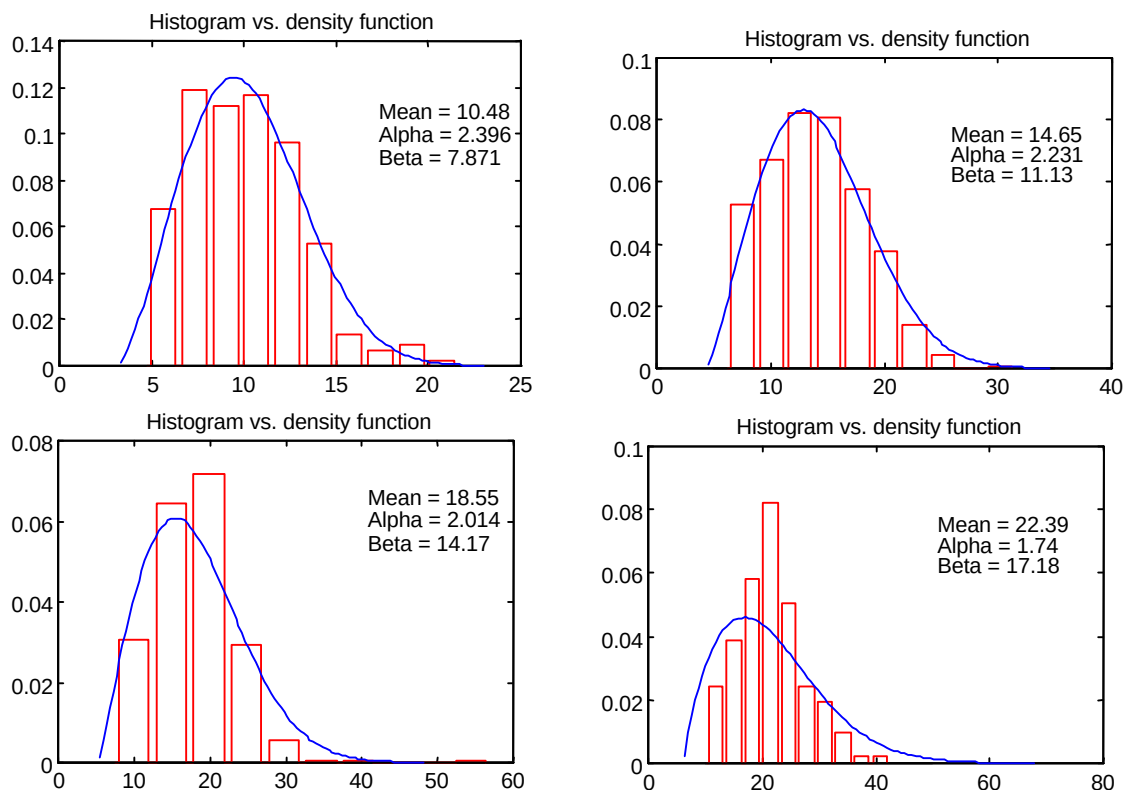


Figure 8.—Plot of histogram vs. distribution for the 2, 6, 10, 14, 18, and 22 cm diameter classes.

Testing Hypothesis of Weibull Distributions

We ran all of the individual plots through both Kolmogorov Smirnov (KS) and Chi Square hypothesis tests (A FORTRAN program for doing this is available from the authors). At the 95 percent level with the alpha and beta parameters computed from the mean diameter, we would reject the hypothesis that the distribution observed comes from the Weibull distribution, for only 32 of the 286 combinations of plot and year of remeasurement using any one of the tests. More details are shown in Table 3 below. Not all plots rejected by the KS test were rejected by the Chi Square test and vice versa. Only 10 plots can be rejected at the 99 percent level. Since the distributions were developed from the aggregated diameter classes, this would appear to be quite reasonable confirmation of the validity of the approach.

Table 3.—Number of rejected plots using the Kolmogorov Smirnov and Chi Square tests at various levels on the 283 control plots

	283 Control Plots Hypothesis Test Level (%)			1122 Permanent Sample Plots Hypothesis Test Level (%)		
	95	97.5	99	95	97.5	99
Number of KS Rejects	13	6	3	63	42	29
Number of chi-square Rejects	27	16	10	112	71	47

Even if all of the plots really were from our Weibull distributions, we would have expected to see 14 (283x5 percent) and 3 (283x1 percent) rejects at the 95 percent and 99 percent levels respectively. Thus, these are quite

encouraging results. The problem plots are confined to 20 of the 80 physical plots. Of the total 32 rejected plot-measurement combinations, 18 of these can be accounted for by 6 plots (561, 2911, 961, 2271, 3811, and 3671). Moreover, as we discuss below, in looking at some of these plots, we can question whether they should be considered typical of plots that we should use to construct normal yield tables (even-aged, fully-stocked stands).

Some Problem Plots and Good Plots

We plotted the histograms for some of the problem plots. These include the problem plots 961, 3671, and 1711. In doing so, we have preprocessed the raw data using the test of insignificant basal area discussed above. Any such plots are given a diameter of 0, and thus appear in the histogram on the extreme left-hand column. This column may also include dead trees since these are assigned a diameter of 0 in the plot measurements.

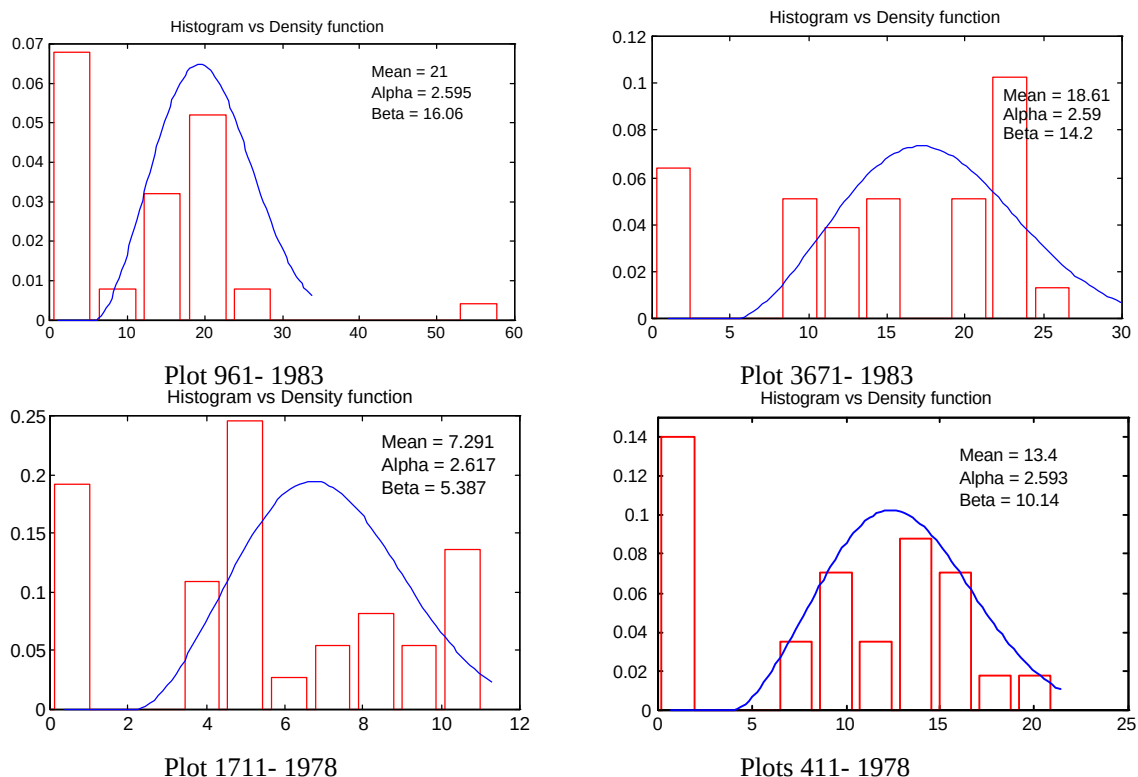


Figure 10.—Histograms of Some Plot Data and the Fitted Distribution.

Plot 961 had one tree of 58 cm in diameter in 1983. This tree can probably account for the rejection of the hypothesis. Plot 3671 is an example of a two-story stand measured in 1983. Plot 1711 in 1978 had a pronounced two-story structure. Although this still remains evident in 1983 and 1988, it is not strong enough in these later years to reject the Weibull hypothesis.

We selected plot 411 in 1978 to give some feel for the quality of fit for those plots not rejected by our hypothesis tests. Although a perfect fit cannot be expected, the degree of correspondence appears quite encouraging.

OTHER SITUATIONS: MANAGED STANDS

We have applied the ideas in this paper to a number of different situations including managed hardwoods and managed and unmanaged softwoods. The essential ideas appear to work well in all these cases, although there are

some differences in the functional form of some of the models. These are reported elsewhere. One interesting point that we found is that there is so little difference in the diameter distributions between managed and unmanaged hardwoods that it is not unreasonable to use the same diameter distribution for both. In Table 3, we show the results of the hypothesis tests for both the 283 control plots used for the normal yield tables and the 1122 hardwood permanent sample plots that the Province maintains to investigate the effect of thinning and cleaning. As can be seen there, the percentage of rejected plots is virtually unchanged. These results were surprising to us initially. However, using a simple simulation model, we have been able to offer at least a partial explanation (Gunn et al. 1997).

GENERAL CONCLUSIONS

In trying to model distributions, we do not have available a simple optimization principle equivalent to the least squares fit that underlies conventional regression modeling. Instead, the onus is on the modeler to choose a "reasonable distribution" with desirable properties, and then do hypothesis testing to establish whether it is defensible to persist in the belief that the distribution is reasonable in face of the data.

In the work reported here, we began with the belief that the Weibull is a reasonable class of distributions to use based on the long history of use of this family of distributions in modeling tree diameters. Our next step was one of aggregating the plot data into mean diameter classes and then calculating MLE estimates of the parameters of the Weibull distribution for each class. Proceeding on the belief that the parameters should vary smoothly from class to class, we fit low order curves through these parameter estimates to get continuous functions that give all three parameters (λ , α , β) as a function of stand average diameter. Note that other approaches are possible. For example, we could have calculated (MLE) estimates for the three parameters for every individual plot and then attempted to fit regression curves to this much larger (and probably more variable) set of points.

In comparing our density functions to the histograms of the class aggregate data, the correspondence is quite good. Even the visual test provides ample evidence of this. Some might argue that this good correspondence between the class aggregate data and the density functions that we have derived is all the verification that is necessary. In arguing this, one would take the view that class aggregates are more likely to give a clearer picture of province-wide average results than the results on any individual plot, and that the individual plots are not really relevant.

Although not necessarily disagreeing with the above viewpoint, we took the position that it would at least be interesting to learn the proportion of plots for which we would be forced to reject the hypothesis that the trees on that plot had diameters drawn from our calculated distribution (given the average diameter of the plot under consideration). We have tested this hypothesis for every plot using two different tests. In both cases, we conducted the tests under the assumption that we were given all parameters of the distribution. This assumption can be questioned in two regards. First, each plot did play a small part in estimating the parameters of the class within which that plot fell. Second, the trees on the plot are used to compute the plot average diameter, which in turn leads to the calculation of α , β , and λ . Given that this assumption is valid, then we obtained the result that we reject the hypothesis that the plot data corresponds to our distribution only for quite a small proportion of cases, even at the 95 percent level. As a result we can conclude that our approach does provide a not unreasonable method for estimating diameter distributions for the type of stands represented by our data.

DISCLAIMER

The main purpose of this paper is to report on the process of developing the diameter distributions, not the results. The current recommended approach of the Nova Scotia Department of Natural Resources may have changed due to either updated data or a change of thinking within the Department. Anyone wishing information on Nova Scotia's hardwood normal yield tables should consult the Forestry Branch, N.S. Department of Natural Resources, P.O. Box 68, Truro, Nova Scotia B2N 5B8.

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